

Motivation

The modality gap in shared audio-text space can bottleneck Audio-LLM systems.

- Can we mitigate modality gap without scaling audio-text supervision?

Key Idea

Generate text-like embeddings from audio embeddings via diffusion!

- We formulate modality bridging as conditional representation generation.
- Diffusion-Link generates text-like embedding from an audio embedding, which can be used as a condition for downstream LLM decoding.

Method

Setup

Denote audio-text embeddings from a multimodal encoder :

$$(\mathbf{e}_0^a, \mathbf{e}_0^t) \in \mathbb{R}^{d \times d}$$

Forward Process

Apply the same forward process to each modality $\mathcal{M} \in \{a, t\}$:

$$\mathbf{e}_s^{\mathcal{M}} = \sqrt{\alpha_s} \mathbf{e}_0^{\mathcal{M}} + \sqrt{1 - \alpha_s} \epsilon_{\mathcal{M}}, \quad \epsilon_{\mathcal{M}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

Training Objective

The denoiser $\phi_{\theta}(\cdot, s)$ is trained to map both noised text and audio embeddings to the text embedding distribution.

- Cross-sample Prediction Loss :

$$\mathcal{L}_{\text{diff}} = \mathbb{E}[\underbrace{\|\mathbf{e}_0^t - \phi_{\theta}(\mathbf{e}_s^t, s)\|_2^2}_{\text{text} \rightarrow \text{text}} + \underbrace{\|\mathbf{e}_0^t - \phi_{\theta}(\mathbf{e}_s^a, s)\|_2^2}_{\text{audio} \rightarrow \text{text}}]$$

- Batch-level Topology Loss :

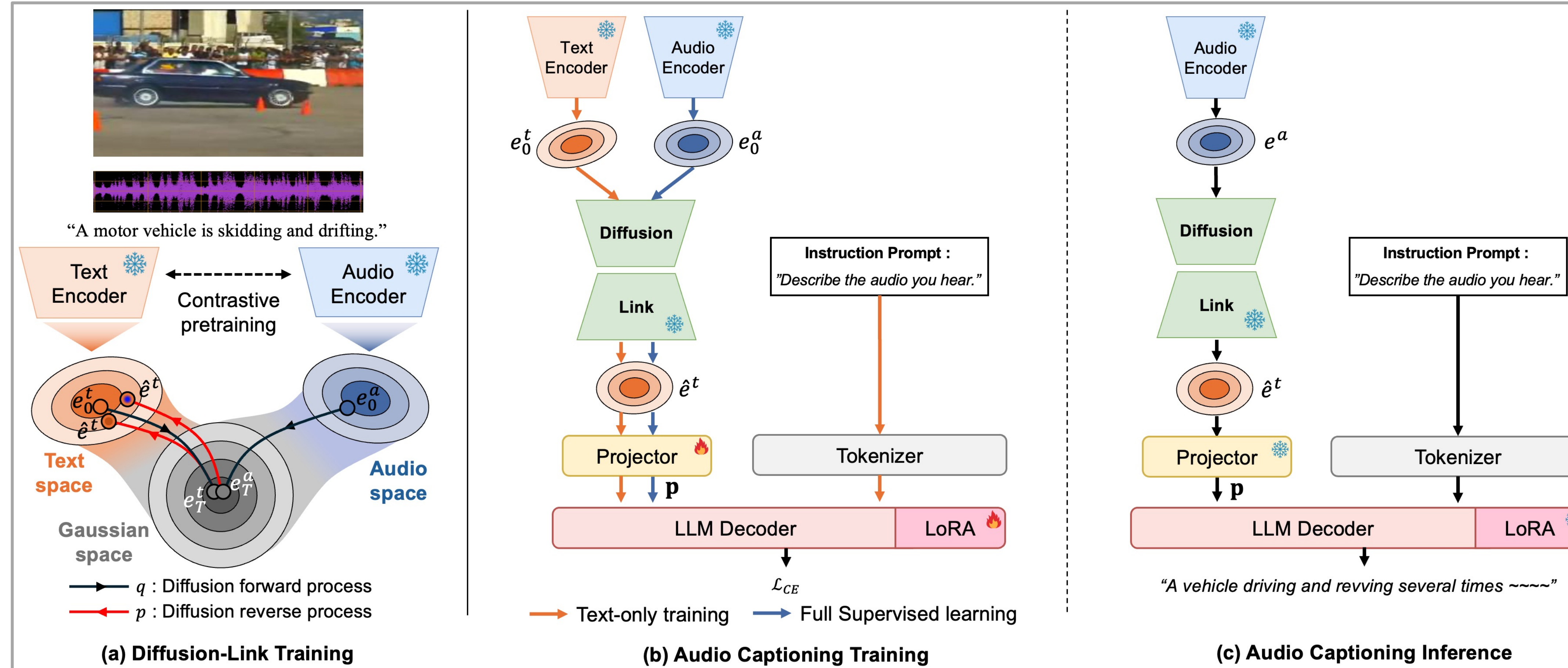
$$\mathcal{L}_{\text{topo}} = \|\mathbf{S}_{xx} - \mathbf{S}_{x\hat{x}}\|_F^2$$

Inference & LLM-based Text Decoding

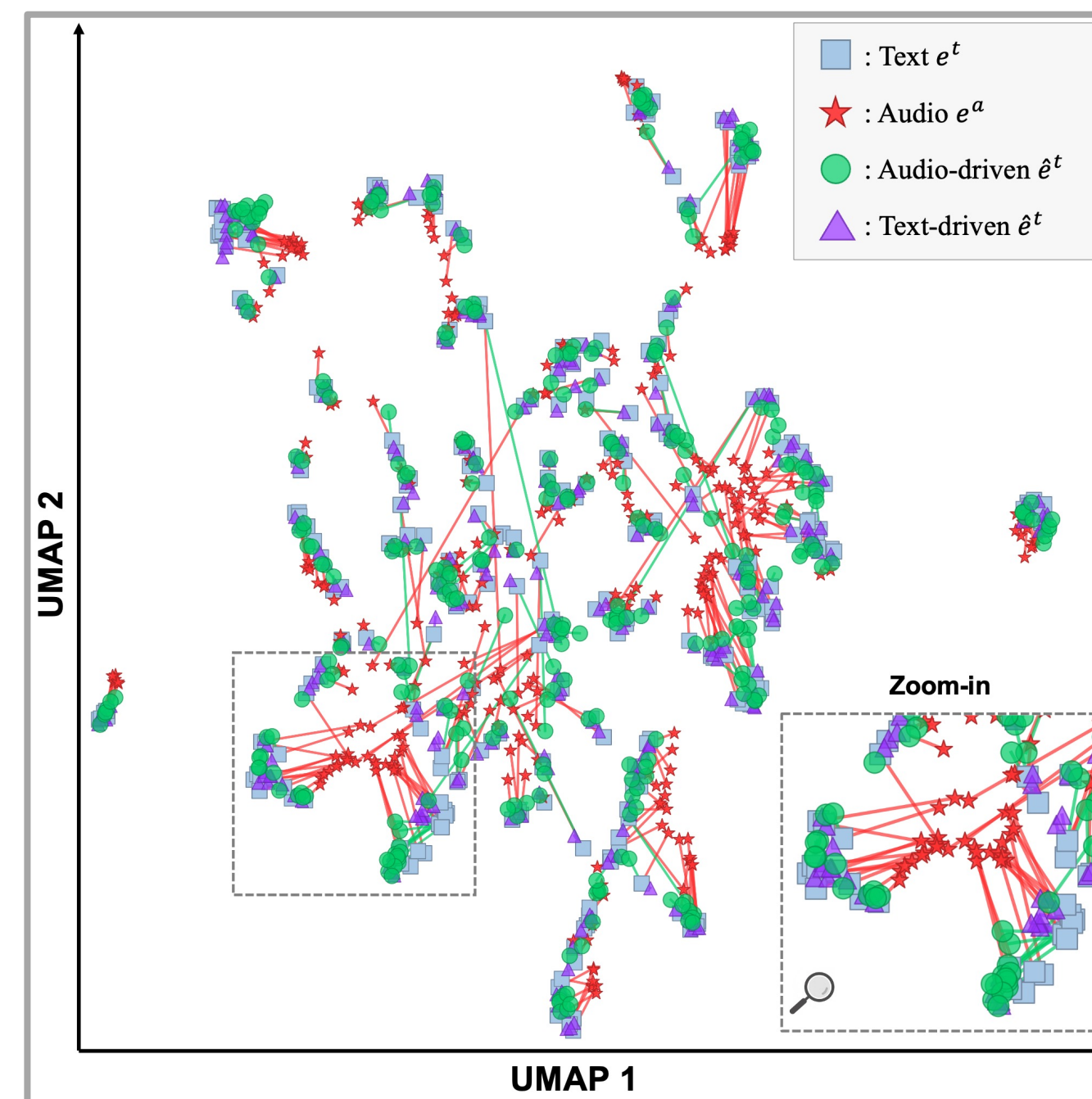
$$\mathbf{e}_{s_*}^{\mathcal{M}} = \sqrt{\alpha_{s_*}} \mathbf{e}^{\mathcal{M}} + \sqrt{1 - \alpha_{s_*}} \epsilon \quad (\text{Forward})$$

$$\hat{\mathbf{e}}^t = \text{DDIM}(\phi_{\theta}, \mathbf{e}_{s_*}^{\mathcal{M}}, s_* \rightarrow 0) \quad (\text{Reverse})$$

Our Framework



Main Results



Method	Output Dim	Ext. Knowledge #	ME $\uparrow$	CD $\uparrow$	SP $\uparrow$	SD $\uparrow$
Zero-shot Captioning						
ZerAuCap	1×D	527	12.3	28.1	8.6	18.3
DRCap	1×D	450k	21.8	59.5	15.7	37.6
Zhang et al.	1×D	No	22.0	64.4	15.6	40.0
WSAC	1×D	46k	24.1	63.3	17.3	40.3
Ours	1×D	No	24.2	73.2	17.5	45.4
Fully Supervised Captioning						
Prefix AAC	T×D	No	24.0	73.3	17.7	45.5
RECAP	T×D	600k	25.6	75.1	18.6	47.1
EnCLAP-large	T×D	No	25.5	80.3	18.8	49.5
CLAP-ART	T×D	No	25.6	80.7	18.8	49.8
Ours	1×D	No	25.6	82.5	18.9	50.7

+41.6%

matched cosine sim
vs. CLAP (matched pairs)

+52.5%

relative CIDEr
zero-shot AAC

+7.3%

relative CIDEr
supervised AAC

- Diffusion-Link improves matched audio–text similarity by **+41.6%**, indicating reduced residual modality gap.
- Without retrieval-based(RAG) external knowledge, it achieves SOTA zero-shot AAC and improves by **+52.5%**.
- Under fully supervised AAC, it still improves by **+7.3%** over the baseline.
- The improvement comes from a **lightweight bridge**: frozen audio-language encoder + 3-block residual MLP diffusion module.

Ablation

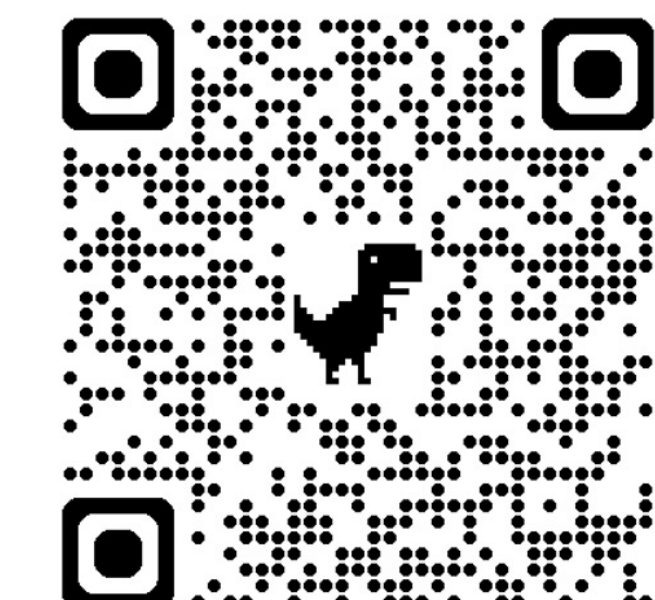
Method	ME $\uparrow$	CD $\uparrow$	SP $\uparrow$	SD $\uparrow$
Zero-shot Captioning				
Baseline (CLAP + LLaMA2-7B)	21.2	48.0	14.4	31.2
+ Diffusion-Bridge	23.3	62.6	16.5	39.5
+ Diffusion-Link	24.2	73.2	17.5	45.4
Fully Supervised Captioning				
Baseline (CLAP + LLaMA2-7B)	25.0	76.9	18.6	47.7
+ Diffusion-Bridge	25.2	77.1	18.0	47.4
+ Diffusion-Link	25.6	82.5	18.9	50.7

Comparison Pair	Cosine Similarity				
	CLAP	C3	DB	DG	DL
(a) $\mathbf{e}^t \cdot \hat{\mathbf{e}}^{t \leftarrow a}(\sim)$	0.486	0.547	0.528	0.110	0.688
(b) $\mathbf{e}^t \cdot \hat{\mathbf{e}}^{t \leftarrow t}(\sim)$	1.000	1.000	0.999	0.334	0.945
(c) $\mathbf{e}^t \cdot \hat{\mathbf{e}}^{t \leftarrow a}(\not\sim)$	0.030	0.092	0.000	0.007	0.000
(d) $\mathbf{e}^t \cdot \hat{\mathbf{e}}^{t \leftarrow t}(\not\sim)$	0.098	0.158	0.002	0.043	0.001

Conclusion

- Residual modality gap in shared audio–text space can bottleneck effective multimodal encoder–LLM coupling.
- Diffusion-Link bridges this gap by generating text-like embeddings toward the text-embedding distribution.**
- The method is **lightweight and plug-in**: the multimodal encoder remains frozen, and the bridge is only a **3-block residual MLP diffusion module**.
- Beyond AAC, Diffusion-Link can extend to audio QA, audio reasoning, and instruction-following Audio-LLMs.
- Future direction: extend Diffusion-Link into a generalized omni-modal generative bridge for connecting representation spaces across modalities.**

Our code



<https://github.com/kaistmm/Diffusion-Link>

Our paper

